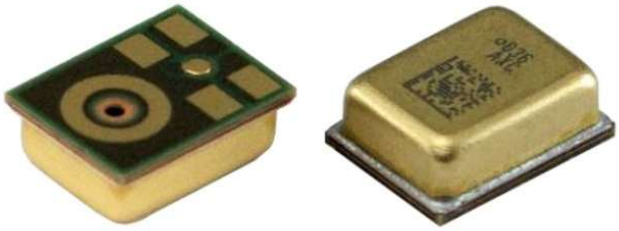
MM07-62964-B66

MEMS Microphone Pair

The MM07 is a miniature, high performance, low power, bottom port silicon microphone.

Product Features

- Small Package / Bottom Port
- Low Current
- Maximum RF Protection
- Ultra-Stable Performance
- Standard SMD Reflow
- Omnidirectional



© Copyright 2023 Knowles

Acoustic & Electrical Specifications

Table 1: General Microphone Specifications
Test Conditions: 21 ± 2°C, 50 ± 5% R.H., V_{DD} =1.8V and no load unless otherwise indicated

Parameter		Minimum	Nominal	Maximum	Units	
Supply Voltage (V_{DD})		1.5	1.8	3.6	V	operating range
Sensitivity	@30Hz	-	-3.0	-	dB	re sensitivity @ 1kHz
	@1kHz	-38	-35	-32	dB	dB re 1V / Pa
	@10kHz	-	1.5	-	dB	re sensitivity @ 1kHz
Resonance Peak	Frequency	-	25.5	-	kHz	-
	Amplitude	-	13.0	-	dB	re sensitivity @ 1kHz
Supply Current (Drain)		-	47	52	µA	V_{DD} = 1.8V
Input-Referred Noise, A-Weighted		-	27	-	dB SPL	20 Hz to 20kHz bandwidth
Signal to Noise Ratio, A-Weighted		65	67	-	dB A	20 Hz to 20kHz bandwidth
Humidity Coefficient		-	0.0	-	dB	per %RH
Storage Temperature		-40	-	+100	ºC	-
Operating Temperature		-17	-	+63	ºC	-
Directivity		Omnidirectional				-
Polarity		Increasing output voltage				Increasing pressure at inlet

Functional operation at these or any other conditions beyond those indicated under “Acoustic & Electrical Specifications” is not implied. Exposure beyond those indicated under “Acoustic & Electrical Specifications” for extended periods may affect device reliability.

© Copyright 2023 Knowles

Frequency Response

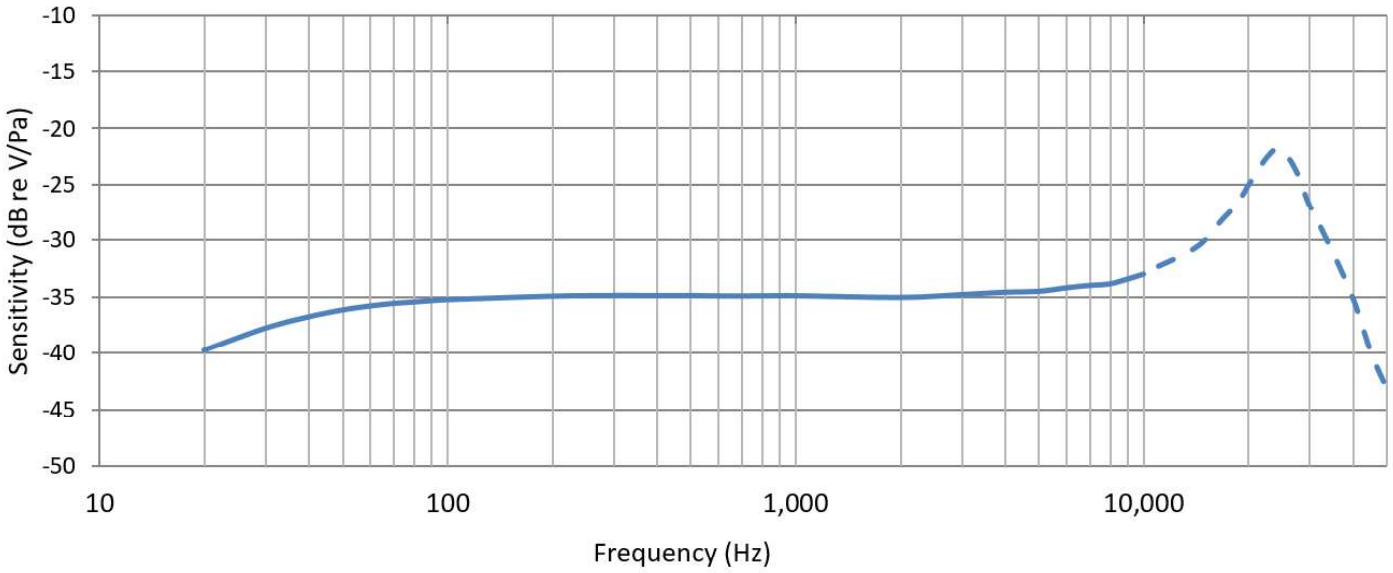


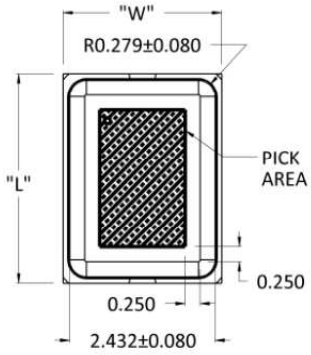
Figure 1: Typical frequency response measured in a pressure cavity under nominal conditions (Table 1). Response above 10 kHz is typical in free field and not measured in the pressure cavity.

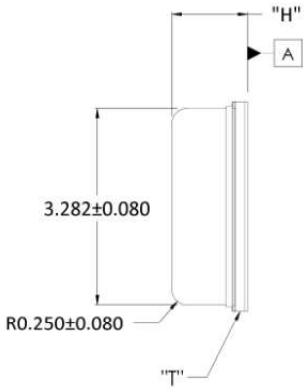
Table 2: Matched Pair Microphone Specifications
Test Conditions: 22 ± 2°C, 50 ± 5% R.H., V_{DD} =1.8V

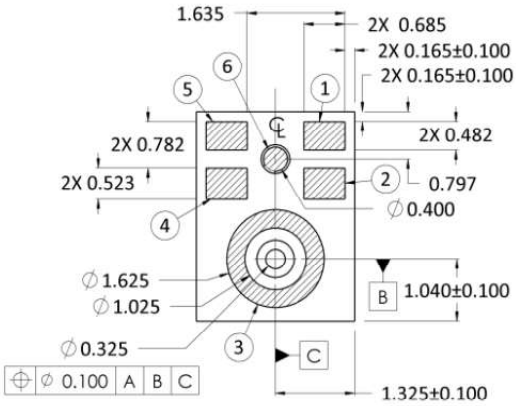
Matching Criteria		Maximum deviation between microphones	Units	
Phase	250 Hz	I 1.5 I	degrees	Microphone with greater phase response value at 250Hz is marked with black stripe.
Sensitivity	1k Hz	I 1.5 I	dB	

© Copyright 2023 Knowles

Mechanical Specifications







Item	Dimension	Tolerance
Length (L)	3.50	±0.10
Width (W)	2.65	±0.10
Height (H)	0.90	±0.10
Acoustic Port (AP)	Ø0.325	±0.05
PCB Thickness (T)	0.25	±0.05

Pin #	Pin Name	Type	Description
1	OUT	Signal	Output
2	GROUND	Power	Ground
3	GROUND	Power	Ground
4	GROUND	Power	Ground
5	Vdd	Power	Power Supply
6	No Connection		

Notes: Pick Area only extends to 0.25 mm of any edge or hole unless otherwise specified.

Dimensions in millimeters [inches]. Not to scale

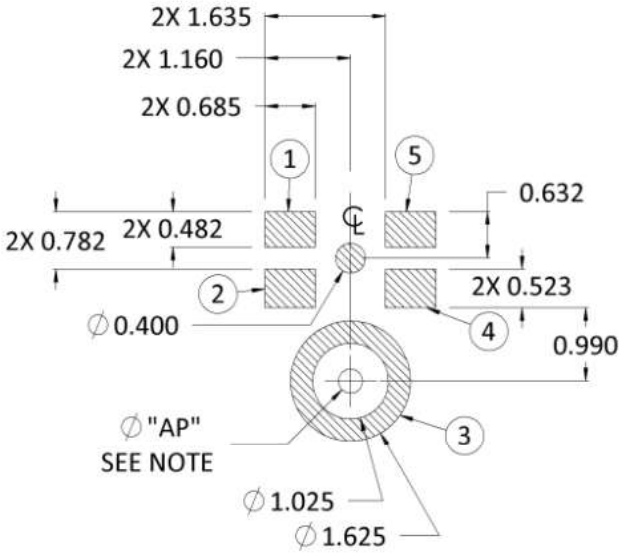
Reflowed cup connected to negative terminal

Weight: 0.024gm

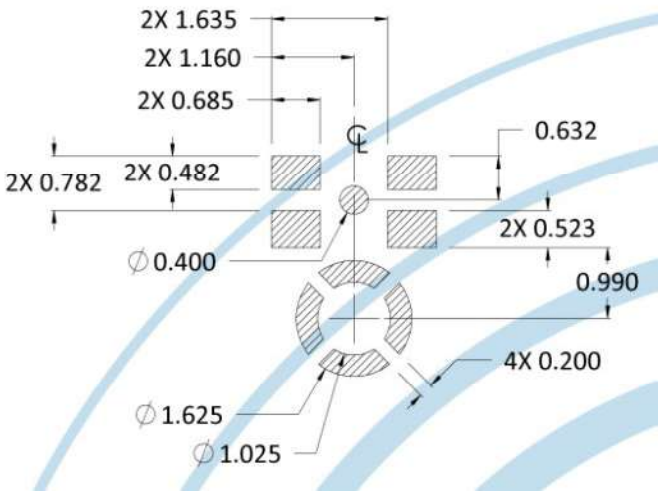
Port location 12Jn

© Copyright 2023 Knowles

Example Land Pattern



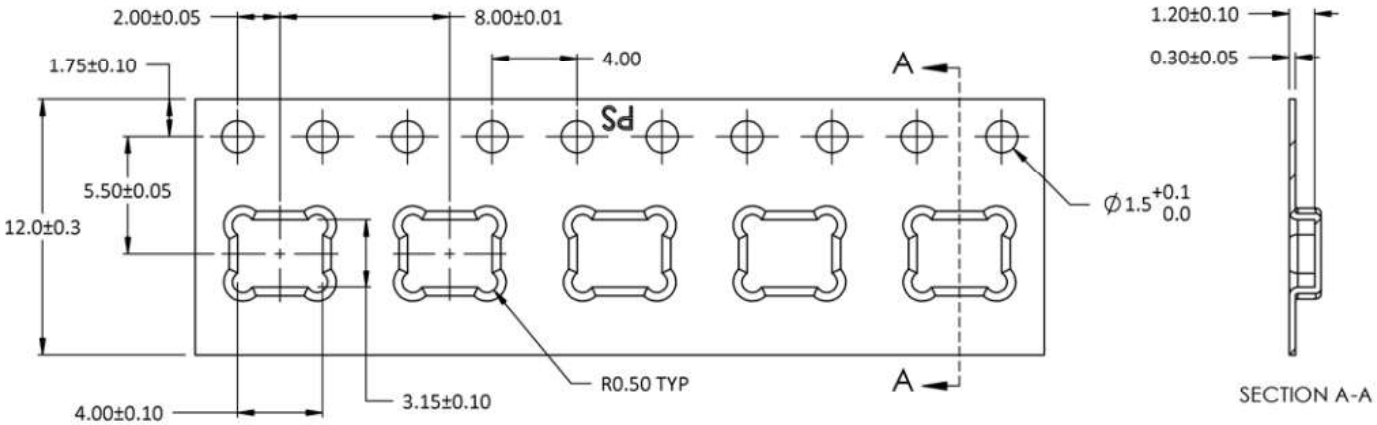
Example Solder Stencil Pattern



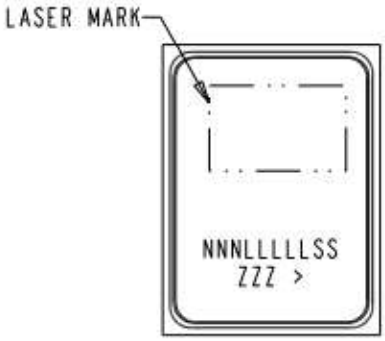
Notes: Dimensions are in millimeters unless otherwise specified.
Further optimizations based on application should be performed.

© Copyright 2023 Knowles

Packaging & Marking Detail



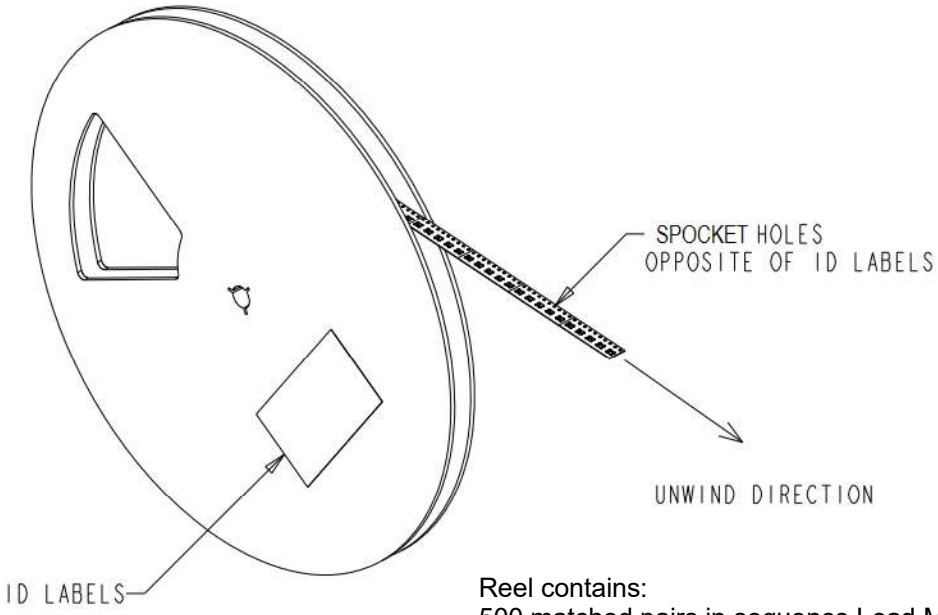
Symbol	Definition	# Characters
N	Part Number (Last 3 digit)	3 (051)
L	Last five characters of Oracle Lot Number	5
S	Sub-Lot Number (01-99)	2
Z	Serial Number	4
>	Lead/Lag symbol (for pairs)	



Microphone with greater phase response value at 250Hz is marked with blue stripe. (Irrespective of the presence of ">" laser marking and serial number)

© Copyright 2023 Knowles

PACKAGING:

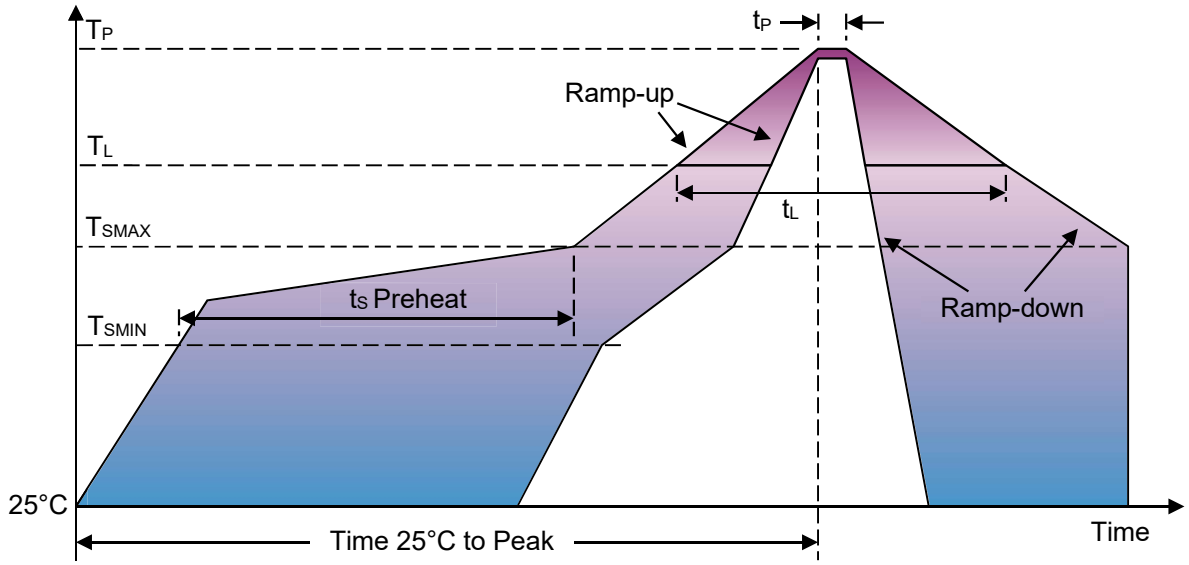


Reel contains:
500 matched pairs in sequence Lead Mic, Lag Mic, Empty Pocket, Lead Mic …
60 empty pockets at the start and end.

Notes: Dimensions are in millimeters unless otherwise specified.
Vacuum pickup only in the pick area indicated in Mechanical Specifications.
Tape & reel per EIA-481.
Labels applied directly to reel and external package.
Shelf life: Twelve (12) months when devices are to be stored in factory supplied, unopened ESD moisture sensitive bag under maximum environmental conditions of 30°C.

© Copyright 2023 Knowles

Recommended Reflow Profile



Profile Feature	Pb-Free
Ramp-up rate (25°C to T_{SMIN})	3°C/second (see note)
Average Ramp-up rate (T_{SMAX} to T_P)	3°C/second max.
Preheat • Temperature Min (T_{SMIN}) • Temperature Max (T_{SMAX}) • Time (T_{SMIN} to T_{SMAX}) (t_S)	 150°C 200°C 60-180 seconds
Time maintained above: • Temperature (T_L) • Time (t_L)	 217°C 60-150 seconds
Peak Temperature (T_P)	260°C
Time within 5°C of actual Peak Temperature (t_P)	20-40 seconds
Ramp-down rate (T_P to T_{SMAX})	6°C/second max
Time 25°C to Peak Temperature	8 minutes max

Notes: Based on IPC/JDEC J-STD-020 Revision C.
All temperatures refer to topside of the package, measured on the package body surface
Ramp-up rate (25°C to T_{SMIN}) can be optimized based on application.

Additional Notes

(A) MSL (moisture sensitivity level) Class 1.
(B) Maximum of 3 reflow cycles is recommended.
(C) In order to minimize device damage:
- Do not board wash or clean after the reflow process.
- Do not brush board with or without solvents after the reflow process.
- Do not directly expose to ultrasonic processing, welding, or cleaning.
- Do not insert any object in port hole of device at any time.
- Do not apply over 30 psi of air pressure into the port hole.
- Do not pull a vacuum over port hole of the microphone.
- Do not apply a vacuum when repacking into sealed bags at a rate faster than 0.5 atm/sec.

.

© Copyright 2023 Knowles

Materials Statement

Meets the requirements of the European RoHS directive 2011/65/EC as amended.

Meets the requirements of the industry standard IEC 61249-2-21:2003 for halogenated substances and Knowles Green Materials Standards Policy section on Halogen-Free.

Product is Beryllium Free according to limits specified on the Knowles Hazardous Material List (HSL for Products).

Ozone depleting substances are not used in the product or the processes used to make the product, including compounds listed in Annex A, B, and C of the “Montreal Protocol on Substances That Deplete the Ozone Layer.”

This product meets RoHS and REACH requirements as of date of publication. To receive latest certificate of compliance, send email to environmental@knowles.com

Reliability Specifications

Test	Description
Reflow	5 reflow cycles with peak temperature of +260°C
High Temperature Storage	+105°C environment for 1,000 hours (IEC 68-2-2 Test Ba)
Low Temperature Storage	-40°C environment for 1,000 hours (IEC 68-2-1 Test Aa)
High Temperature Bias	+105°C environment while under bias for 1,000 hours (IEC 68-2-2 Test Ba)
Low Temperature Bias	-40°C environment while under bias for 1,000 hours (IEC 68-2-1 Test Aa)
Temperature/Humidity Bias	+85°C/85% R.H. environment while under bias for 1,000 hours (JESD22-A101A-B)
Thermal Shock	100 cycles of air-air thermal shock from -40°C to +125°C with 15 minute soaks (IEC 68-2-4)
Vibration	16 minutes in each X, Y, Z axis from 20 to 2,000 Hz with peak acceleration of 20 G (MIL 883E, Method 2007.2,A)
Mechanical Shock	3 pulses of 10,000 G in each of the X, Y, and Z directions (IEC 68-2-27 Test Ea)
ESD-HBM	3 discharges of ±2kV direct contact to I/O pins (MIL 883E, Method 3015.7)

Notes: After reliability tests are performed, the sensitivity of the microphones shall not deviate more than 3dB from their initial value.
After 3 reflow cycles, the sensitivity of the microphones shall not deviate more than 1 dB from its initial value.

© Copyright 2023 Knowles

Specification Revisions

Revision	Specification Changes	Date
A	Design	06-09-23

Date of Publication:

Information contained herein is subject to change without notice. It may be used by a party at their own discretion and risk. We do not guarantee any results or assume any liability in connection with its use. This publication is not to be taken as a license to operate under any existing patents.

Knowles Corporation
1151 Maplewood Drive
Itasca, Illinois 60143

Phone: (630) 250-5100
Fax: (630) 250-0575
sales@knowles.com

knowles.com

© Copyright 2023 Knowles

© 2023, Knowles Electronics, LLC, Itasca, IL USA, – All Rights Reserved. Knowles and the logo are trademarks of Knowles Electronics, LLC.